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RESEARCH ARTICLE

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Clinical characteristics and prognosis of IgA nephropathy and comorbid diabetic kidney disease

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ABSTRACT

Aims: This study characterized clinical features, pathology, treatment, and prognosis of biopsy-confirmed coexisting IgA nephropathy (IgAN) and diabetic kidney disease (DKD).

Methods: This single-centre retrospective study included 80 patients with biopsy-proven IgAN + DKD (2010–2023). Propensity score matching (PSM) (1:1) was used to establish comparable groups of IgAN (n = 80) and DKD (n = 80), matched on age, sex and baseline serum creatinine (sCr). Baseline clinical and pathological data, treatments, 1-year response and long-term outcomes were compared.

Results: In the unmatched cohort, IgAN + DKD patients had higher baseline sCr, greater 24-hour urinary protein (24h-uPro), and less microscopic haematuria (MH) than IgAN. After matching, differences in 24h-uPro and MH persisted. IgAN + DKD showed fewer crescents, lower KM55/IgA ratio, and weaker C3 deposition. Versus DKD, IgAN + DKD had more MH but milder glomerular lesions and arteriolar hyalinosis. In patients with 1-year data (n = 31), 24h-uPro of IgAN+DKD declined substantially, with 58.06% overall clinical remission. Median time free of the composite kidney endpoint (≥50% estimated glomerular filtration rate (eGFR) decline or kidney failure) was 65.63 months; risk was higher than IgAN (HR 2.821, 95% CI 1.174–6.781) but lower than DKD (HR 0.525, 95% CI 0.293–0.943). Baseline sCr and irreversible GBM thickening associated with endpoints. Unselected immunosuppression showed no long-term renal benefit.

Conclusions: IgAN + DKD exhibited lower intrarenal immune activity than IgAN and less severe diabetic glomerular injury than DKD. The KM55/IgA ratio might aid differentiation from IgAN. Long-term renal prognosis was intermediate between the monodisease groups. Routine immunosuppression reduced short-term proteinuria but did not improve hard renal endpoints. Higher baseline sCr and GBM thickness linked to poorer outcomes.

KEY MESSAGES

- The KM55/IgA immunofluorescence ratio may serve as a potential biomarker to distinguish IgAN + DKD from IgAN.
- Unselected immunosuppressive therapy is not associated with measurable long-term renal benefit in IgAN + DKD.

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IgAN; DKD; clinical characteristics; therapeutic approaches; prognosis

1. Introduction

IgA nephropathy (IgAN) represents the most prevalent form of primary glomerulonephritis and constitutes one of the principal causes of renal failure [1]. A recent large-scale retrospective cohort study showed the median time to renal failure for IgAN is approximately 10.7 years, and the 10-year renal survival rate of 0.52 [2]. IgAN with other comorbid glomerular diseases is termed complex IgAN, with the primary concurrent pathological types being diabetic kidney disease (DKD) (46.25%) and membranous nephropathy (17.5%) [3]. DKD is a prevalent secondary kidney disease and also a major

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contributor to kidney failure [4]. It has been reported that one-third of diabetic patients suffered from both DKD and non-DKD renal lesions, and the superimposition of IgAN in DKD patients is common [5,6].

IgAN and DKD have distinct pathogeneses and clinical features. IgAN is an immune complex-mediated inflammatory glomerulopathy. Aberrant glycosylation of IgA1 leading to increased circulating galactose-deficient IgA1 (Gd-IgA1) plays a pivotal role in its pathogenesis, culminating in mesangial deposition of immune complexes and subsequent glomerular injury. Clinically, it typically presents with recurrent gross or persistent microscopic haematuria (MH), varying degrees of proteinuria, hypertension and progressive renal insufficiency [7]. DKD, a major microvascular complication of diabetes, arises from the synergistic effects of chronic hyperglycaemia-induced metabolic derangements and microvascular damage. Its hallmark manifestations include progressively increasing albuminuria/proteinuria and declining renal function, often culminating in end-stage kidney disease [8]. The treatments and prognosis of IgAN and DKD differ substantially. IgAN typically responds favourably to personalized immunosuppressive therapy [9,10], whereas DKD management primarily focuses on non-immunosuppressive interventions, such as glycaemic control, blood pressure regulation and lipid management [11]. The prognosis of DKD is poorer than that of non-DKD renal lesions, with a more significant decline in glomerular filtration rate (GFR) [12]. In patients with concurrent IgAN and DKD, immune and metabolic factors may interact synergistically, potentially leading to clinical phenotypes, treatment responses and progression risks that differ from those in pure IgAN or pure DKD.

While prior biopsy series have documented the prevalence of non-diabetic lesions in diabetic patients, dedicated comparative studies of biopsy-confirmed IgAN + DKD versus pure forms remain limited, particularly regarding prognosis and remission under contemporary care. Our study provides a structured, comparative analysis of clinical characteristics, pathological features, short-term remission and follow-up dynamics in three well-defined groups: IgAN + DKD, pure IgAN and pure DKD. This approach allows for better delineation of how comorbid pathology influences presentation and early outcomes, offering insights that may inform risk stratification, biopsy decision-making and tailored therapy in patients with overlapping glomerular diseases.

2. Methods

2.1. Study population and inclusion criteria

This single-centre retrospective study was approved by the Ethics Committee of Hangzhou Traditional Chinese Medicine Hospital (HZTCMH) (Approval No. 2025KLL010). The requirement for consent was waived by the Ethics Committee. The study had been performed in accordance with the principles stated in the Declaration of Helsinki. It included patients diagnosed with IgAN + DKD via renal biopsy at HZTCMH between 1 January 2010 and 31 December 2023.

Diagnostic criteria:

1. IgAN: defined as dominant or co-dominant mesangial IgA deposition on immunofluorescence (intensity ≥1.5+ on a 0–4 scale), with mesangial proliferative changes on light microscopy (LM) [13].
2. DKD: confirmed type 2 diabetes mellitus (diagnosed by American Diabetes Association criteria), with typical renal histopathologic features, including glomerular basement membrane (GBM) thickening (female >395 nm, male >430 nm), mesangial expansion, nodular sclerosis and/or arteriolar hyalinosis [14].

Inclusion criteria:

1. Renal biopsy was performed, with immunofluorescence confirming that IgA deposition was the dominant finding in the glomerular mesangial region [13].
2. Pathological features were consistent with the diagnostic criteria for DKD [14].
3. Minimum follow-up duration was at least 8 weeks.
4. 24-Hour urinary protein (24h-uPro) and GFR estimated using the EPI equation (eGFR) were necessary both at the time of biopsy and follow-up.

Exclusion criteria:

1. Comorbidity with other primary or secondary kidney diseases.
2. Presence of pre-renal, post-renal or drug/infection-related acute kidney injury.

Patients diagnosed with IgAN and DKD during the same period were enrolled in the control groups (Figure 1). Baseline clinical data, therapies and follow-up outcomes were compared among the three groups.

2.2. Study indicators

Data were extracted from the Electronic Medical Record System. Baseline was defined as the date kidney biopsy was performed. All clinical and laboratory variables at baseline were collected from the closest available measurement within 10 days prior to the time of biopsy.

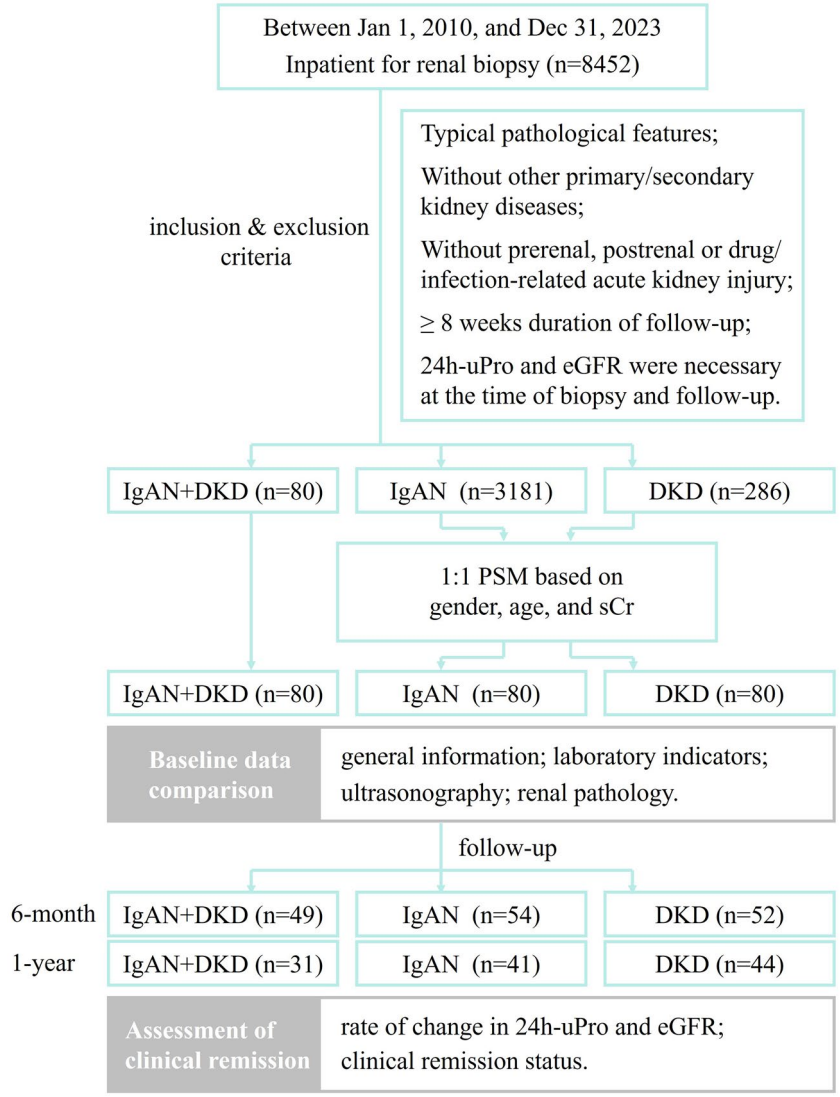


Figure 1. Flowchart of participant selection. Between 2010 and 2023, a total of 8452 patients underwent renal biopsy at our institution. Among them, 80 patients with IgAN + DKD met the predefined inclusion and exclusion criteria, alongside 3181 patients with pure IgAN and 286 patients with pure DKD. Using PSM (1:1, matching factors: sex, age and sCr), 80 patients with pure IgAN and 80 patients with pure DKD were matched to the IgAN + DKD group. Baseline data of the three groups were compared. Clinical remission was evaluated at the 6th month and 1 year. 24h-uPro: 24-hour urinary protein; eGFR: estimated glomerular filtration rate; IgAN: IgA nephropathy; DKD: diabetic kidney disease; PSM: propensity score matching; sCr: serum creatinine.

1. General information: sex, age, body mass index (BMI), diabetes duration, diabetic retinopathy (DR) and hypertension duration.
2. Laboratory indicators: eGFR (using 2021 chronic kidney disease epidemiology (CKD-EPI) creatinine equation, eGFR = $142 \times \min(sCr/\kappa, 1)^{\alpha} \times \max(sCr/\kappa, 1)^{-1.200} \times 0.9938^{Age} \times 1.012$ [if female], sCr = standardized serum creatinine (sCr) in mg/dL, $\kappa = 0.7$ [female] or 0.9 [male], $\alpha = -0.241$ [female] or −0.302 [male], $\min(sCr/\kappa, 1)$ is the minimum of sCr/κ or 1.0, $\max(sCr/\kappa, 1)$ is the maximum of sCr/κ or 1.0, age (years)), sCr, haemoglobin A1c (HbA1c), serum albumin (sAlb), total cholesterol (TC), triglyceride (TG), low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), lipoprotein(a) (Lp(a)), apolipoprotein A (ApoA), apolipoprotein B (ApoB), apolipoprotein E (ApoE); haemoglobin (Hb), serum immunoglobulin A (sIgA), serum immunoglobulin M (sIgM), serum immunoglobulin G (sIgG), serum complement component 1q (sC1q), serum complement component 3 (sC3), serum complement component 4 (sC4); 24h-uPro, urine red blood cell (uRBC). Microscopic haematuria refers to >3 RBC per high-power field on microscopic evaluation of a single properly collected urine specimen.
3. Ultrasonography: interventricular septal thickness at diastole (IVSD), left ventricular internal diastolic dimension (LVIDD), left ventricular internal diameter in systole (LVIDS), left ventricular posterior wall dimensions (LVPWD), aortic root diameter (AO), left atrium (LA) and left ventricular ejection fraction (LVEF).
4. Renal pathology: all renal pathological specimens were evaluated by LM, immunofluorescence and electron microscopy (EM) and diagnosed by two experienced pathologists. Based on the Oxford MEST-C classification and DKD grading criteria [13,14], the severity of glomerular, tubule, interstitial and renal arteriolar lesions was graded. Semi-quantitative scoring was performed based on the immunofluorescence intensity of IgA, IgM, IgG, C3, C4, C1q, κ, λ and KM55 deposition, with scores ranging from 0 to 4+. EM was used to assess the GBM thickness and the fusion of foot processes.

2.3. Treatment, follow-up and composite endpoint

Treatments administered after renal biopsy were documented, including supportive care (low-salt diet, angiotensin-converting enzyme inhibitors/angiotensin II receptor blockers (ACEI/ARB), sodium-glucose transporter 2 inhibitors (SGLT2i)), glucocorticoids and immunosuppressants. The immunosuppressants included cyclophosphamide (CTX), mycophenolate mofetil (MMF), cyclosporine A (CsA), tacrolimus (FK506), leflunomide (LEF) and *Tripterygium wilfordii* polyglycoside tablets (TW).

The minimum follow-up duration was set at 8 weeks to allow at least one post-baseline safety and efficacy evaluation. Patients with shorter follow-up were excluded. The assessment of clinical remission was performed at 12 ± 0.5 months after baseline, while available shorter-term data (including 6 months) were also analysed and reported. Clinical remission was categorized as follows:

Complete response: 24h-uPro ≤0.3 g + sCr increase <10% vs. baseline;
Partial response: 0.3 g < 24h-uPro < 1 g (≥50% reduction) + sCr increase <10% vs. baseline;
Non-response: failure to meet the above criteria for 24h-uPro or sCr increase >10% vs. baseline.
Response rate = complete response rate + partial response rate.

Composite endpoint was defined as a permanent decline in eGFR of ≥50% from baseline or progression to kidney failure (eGFR <15 mL/min/1.73 m^2 or maintenance dialysis). Permanent eGFR decline was adjudicated as confirmed on two consecutive measurements at least 1 month apart (to distinguish from transient fluctuations or acute events), with the second measurement confirming the decline. The date of the first qualifying measurement was used as the event date. If only one measurement showed decline but follow-up was lost, it was not counted as an event (censored instead). The last follow-up and endpoint incidence were recorded.

2.4. Statistical analyses

Data analyses were performed using SPSS26 (Chicago, IL) and R (4.5.1) (Vienna, Austria), along with Zstats v1.0 (www.zstats.net). To reduce the potential bias caused by baseline differences between the

IgAN + DKD and control groups, propensity score matching (PSM) was performed. Propensity scores were calculated using a logistic regression model that included age, sex and sCr level as covariates. Patients in the IgAN + DKD group were matched 1:1 with those in the IgAN and DKD groups based on their propensity scores using a nearest-neighbor matching algorithm without replacement and a caliper width of 0.2 of the standard deviation of the logit of the propensity score. The balance between the matched groups was assessed using standardized mean differences, with a threshold of <0.1 indicating adequate balance. Normality was assessed using the P–P plot combined with the Shapiro–Wilk test. Continuous variable data meeting the normality assumption were expressed as mean ± standard deviation, and intergroup comparisons were performed using one-way analysis of variance (ANOVA) followed by the least significant difference test. Skewness data are presented as medians (P25 and P75), and intergroup comparisons were conducted using nonparametric tests. Categorical data were expressed as counts (percentages), and intergroup comparisons were analysed using Pearson's chi-square test or Fisher's exact test, as appropriate. All time-to-event analyses were performed under the assumption of non-informative censoring. Patients were right-censored at the date of the last follow-up which they were confirmed to be alive and free of the event of interest. The Kaplan–Meier method was used to estimate survival functions and generate survival curves. Intergroup comparisons were performed using the log-rank test. The median survival time and interquartile range for each group were derived from the Kaplan–Meier curves. Cox proportional hazards models were used to assess the association of covariates with time to the composite endpoint in the IgAN + DKD group. Univariate Cox regression was initially performed to screen potential risk factors for the composite endpoint. Variables with $p < 0.05$ in univariate analysis were considered candidates for multivariate analysis. Multivariate Cox regression was performed using a backward stepwise selection method, with removal criterion $p \geq 0.05$. To address potential multicollinearity, highly correlated variables (e.g. eGFR and sCr) were not entered simultaneously, and variance inflation factors (VIFs) were calculated for candidate variables; those with VIF > 5 were considered for exclusion or combination prior to modelling. The proportional hazards assumption was tested using Schoenfeld residuals, and no major violations were observed (global $p > 0.05$). Results are presented as hazard ratios (HRs) with 95% confidence intervals (95%CIs) and corresponding p values. All multivariable Cox regression analyses were predefined as exploratory outcomes to identify potential risk factors, rather than confirmatory hypothesis testing. Missing pattern analysis used Welch's t-test for variables with ≥5% missingness. Little's Missing Completely at Random (MCAR) test was used to statistically assess whether the missing data pattern deviated from the MCAR assumption, to inform subsequent missing data handling strategies.

3. Results

3.1. Baseline characteristics

Eighty patients diagnosed with IgAN + DKD were enrolled in the study. During the same period, 3181 cases of IgAN and 286 cases of DKD were identified, with ratios of 1:40 and 1:3.58, respectively. Compared to the IgAN group, the IgAN + DKD group showed a higher proportion of male patients, an older average age at renal biopsy, a higher baseline median level of sCr, greater 24h-uPro and a lower incidence of MH. There was no difference between the IgAN + DKD and DKD groups in terms of sex, age, sCr and 24h-uPro; however, the incidence of MH was higher in the IgAN + DKD group (Supplementary Table 1). PSM (1:1, matching factors: sex, age and sCr) established control cohorts for 80 IgAN and 80 DKD patients (Table 1).

Compared to the IgAN group, patients in the IgAN + DKD group had a higher BMI, with 12.5% of them being obese (BMI gr30 kg/m^2). Patients with IgAN + DKD exhibited a longer duration of hypertension. The 24h-uPro (2.31 g vs. 1.09 g, $p < 0.001$) was higher, and 37.5% of the patients in the IgAN + DKD group had a 24h-uPro ≥3.5 g. In contrast, the incidence of MH (57.50% vs. 77.50%, $p = 0.007$), uRBC count and proportion of dysmorphic uRBC were lower in the IgAN + DKD group than in the IgAN group. Notably, abnormalities in lipid metabolism were detected in the IgAN group; the levels of TC, TG and LDL-C were higher, whereas HDL-C, ApoA and ApoE levels were lower than those in the IgAN + DKD group. Echocardiographic findings revealed that LA, IVSD, LVIDD and LVIDS were higher in the IgAN + DKD group, indicating a

Table 1. Baseline clinical data of the IgAN + DKD group, IgAN group and DKD group.

	IgAN + DKD (n = 80)	IgAN (n = 80)	DKD (n = 80)	p (IgAN + DKD vs. IgAN)	p (IgAN + DKD vs. DKD)
Male, n (%)	58 (72.50)	59 (73.75)	57 (71.25)	0.858	0.860
Age (years)	53.86 ± 10.80	53.13 ± 10.18	52.15 ± 9.43	0.769	0.362
BMI (kg/m^2)	25.80 ± 3.79	24.38 ± 2.92	24.87 ± 3.03	0.009	0.089
Diabetes duration (months)	78 (36, 120)	0 (0, 0)	90 (24, 150)	<0.001	0.423
DR, n (%)	24 (30.00)	8 (10.00)	44 (55.00)	0.002	0.001
Hypertension duration (months)	90 (3.75, 183)	4.5 (0, 96)	24 (1, 93)	<0.001	0.006
sCr (μmol/L)	118 (85, 150.25)	107 (86.25, 146)	103.50 (74, 138.50)	0.952	0.232
eGFR (mL/min/1.73 m^2)	66.35 ± 23.44	64.94 ± 23.01	67.63 ± 24.96	0.701	0.737
CKD1–2, n (%)	40 (50.00)	41 (51.20)	44 (55.00)	0.804	0.808
CKD3, n (%)	34 (42.50)	31 (38.80)	30 (37.50)		
CKD4–5, n (%)	6 (7.50)	8 (10.00)	6 (7.50)		
HbA1c (%)	7.38 ± 1.48	5.81 ± 0.64	7.62 ± 2.10	<0.001	0.412
sAlb (g/L)	36.30 (33.40, 39.80)	36.25 (33.53, 39.45)	36.15 (31.33, 38.95)	0.844	0.359
TC (mmol/L)	4.93 (4.11, 5.85)	6.01 (4.91, 6.63)	4.53 (3.89, 5.46)	<0.001	0.171
TG (mmol/L)	1.75 (1.07, 2.75)	2.48 (1.72, 3.33)	1.63 (1.09, 3.05)	0.001	0.902
LDL-C (mmol/L)	3.03 (2.41, 3.73)	3.53 (2.89, 4.38)	2.78 (2.11, 3.31)	0.003	0.141
HDL-C (mmol/L)	1.04 (0.86, 1.23)	0.87 (0.75, 0.95)	1.01 (0.83, 1.20)	<0.001	0.618
Lp(a) (mg/L)	213 (69, 393)	199.45 (74.75, 351.50)	163 (74.50, 370.50)	0.536	0.446
ApoA (g/L)	1.34 (1.19, 1.55)	0.87 (0.75, 0.96)	1.25 (1.11, 1.44)	<0.001	0.056
ApoB (g/L)	0.99 ± 0.24	0.94(0.78, 1.05)	0.93 ± 0.27	0.212	0.256
ApoE (mg/dL)	4.87 (3.88, 6.33)	2.54 (1.75, 3.38)	4.46 (3.66, 5.42)	<0.001	0.390
Hb (g/L)	128.50 (112.50, 144.25)	135 (114.25, 149)	116.50 (102, 136)	0.242	0.004
sIgA (mg/dL)	302.50 (231, 370)	331 (248, 432)	224 (162, 306.50)	0.099	<0.001
sIgM (mg/dL)	83.50 (64, 116.25)	82 (68, 111)	86 (66, 112.50)	0.985	0.665
sIgG (mg/dL)	1086.44 ± 312.25	1072.80 ± 295.40	1068.22 ± 271.42	0.778	0.697
sC1q (mg/L)	203.02 ± 42.05	200.28 ± 38.81	188.80 ± 28.64	0.694	0.009
sC3 (mg/dL)	100 (93, 116)	97 (89.25, 108.75)	99 (92.50, 118.50)	0.095	0.601
sC4 (mg/dL)	26.40 (22, 30.75)	25 (21.25, 28.75)	26 (22, 31)	0.185	0.832
24h-uPro (g)	2.31 (1.30, 4.17)	1.09 (0.56, 2.39)	2.21 (0.92, 4.04)	<0.001	0.477
24h-uPro ≥ 3.5 g, n (%)	30 (37.50)	10 (12.50)	26 (32.50)	<0.001	0.507
MH, n (%)	46 (57.50)	62 (77.50)	25 (31.25)	0.007	0.001
uRBC (/μL)	10 (3, 35.03)	42.30 (11.48, 78)	3 (1, 8.75)	<0.001	<0.001
Dysmorphic uRBC (%)	33.30 (0, 65)	65 (38.13, 70)	0 (0, 48.64)	<0.001	0.002
Echocardiography					
AO (mm)	28.42 ± 4.47	29 ± 3.86	28.84 ± 3.65	0.424	0.947
LA (mm)	35.49 ± 4.86	33.44 ± 4.69	35.43 ± 4.03	0.023	0.964
IVSD (mm)	10.16 ± 1.39	9.64 ± 1.16	10.18 ± 1.62	0.025	0.328
LVIDD (mm)	49.74 ± 4.28	47.50 ± 5.04	48.99 ± 4.49	0.009	0.187
LVIDS (mm)	32.25 ± 5.01	29.66 ± 3.73	31.20 ± 4.46	0.001	0.747
LVPWD (mm)	10 (9, 10.70)	9.50 (9, 10.20)	10 (9, 11)	0.213	0.984
LVEF (%)	65.10 ± 6.47	67.63 ± 5.09	65.99 ± 7.10	0.018	0.466

DR: diabetic retinopathy; sCr: serum creatinine; eGFR: estimated glomerular filtration rate; CKD: chronic kidney disease; HbA1c: haemoglobin A1c; sAlb: serum albumin; TC: total cholesterol; TG: triglyceride; LDL-C: low-density lipoprotein cholesterol; HDL-C: high-density lipoprotein cholesterol; Lp(a): lipoprotein(a); ApoA: apolipoprotein A; ApoB: apolipoprotein B; ApoE: apolipoprotein E; Hb: haemoglobin; sIgA: serum immunoglobulin A; sIgM: serum immunoglobulin M; sIgG: serum immunoglobulin G; sC1q: serum complement component 1q; sC3: serum complement component 3; sC4: serum complement component 4; 24h-uPro: 24-hour urinary protein; MH: microscopic haematuria; uRBC: urine red blood cell; AO: aortic root diameter; LA: left atrium; IVSD: interventricular septal thickness at diastole; LVIDD: left ventricular internal diastolic dimension; LVIDS: left ventricular internal diameter in systole; LVPWD: left ventricular posterior wall dimensions; LVEF: left ventricular ejection fractions.

tendency towards left ventricular enlargement and interventricular septal thickening. Moreover, LVEF was lower in patients with IgAN + DKD, indicating a decline in left ventricular systolic function.

Compared with the DKD group, the IgAN + DKD group had a longer duration of hypertension and no difference in diabetes duration; however, the incidence of DR was lower. There was no difference in 24h-uPro, but the IgAN + DKD group exhibited a higher incidence of MH (57.50% vs. 31.25%, p = 0.001), uRBC count, and proportion of dysmorphic uRBC. No differences were found in the lipid profiles or cardiac structures between the two groups.

3.2. Renal pathology

Compared with the IgAN group, the IgAN + DKD group exhibited more severe mesangial matrix expansion, thicker GBM and a lower proportion of crescent, with no difference in the degree of foot process fusion. A higher proportion of IgAN + DKD patients presented with tubular atrophy exceeding 25%, accompanied by more severe interstitial fibrosis and arteriole sclerosis (Supplementary Tables 2-3, Figure 2(a)). The majority of IgAN + DKD patients showed an IgA fluorescence intensity of 2+ or higher, which

was comparable to that in the IgAN group (Supplementary Table 4). Our previous study demonstrated that the intensity of glomerular anti-human Gd-IgA1 (KM55) deposition and the KM55/IgA ratio were significant for identifying primary IgAN [15]. Thus, we enrolled 40 patients with IgAN using computer-generated randomization. PSM was applied in a 1:2 ratio (matching factors: sex, age, sCr and 24h-uPro). This yielded 16 matched IgAN patients and 28 IgAN + DKD patients. Further analysis of the deposition of KM55 revealed that over 50% of the patients in the IgAN + DKD group had a KM55 fluorescence intensity of 2+ or higher, similar to that in the IgAN group. However, the KM55/IgA ratio was lower (Supplementary Table 5, Figure 2(c)). In addition, the deposition of λ, κ and C3 was more prominent in the IgAN group.

Glomerular lesions in IgAN + DKD patients were less severe than those in DKD patients (Figure 2(b)). Specifically, the IgAN + DKD group exhibited milder mesangial matrix expansion, K–W nodule formation and GBM thickening, while the degree of foot process fusion was more pronounced. There was no

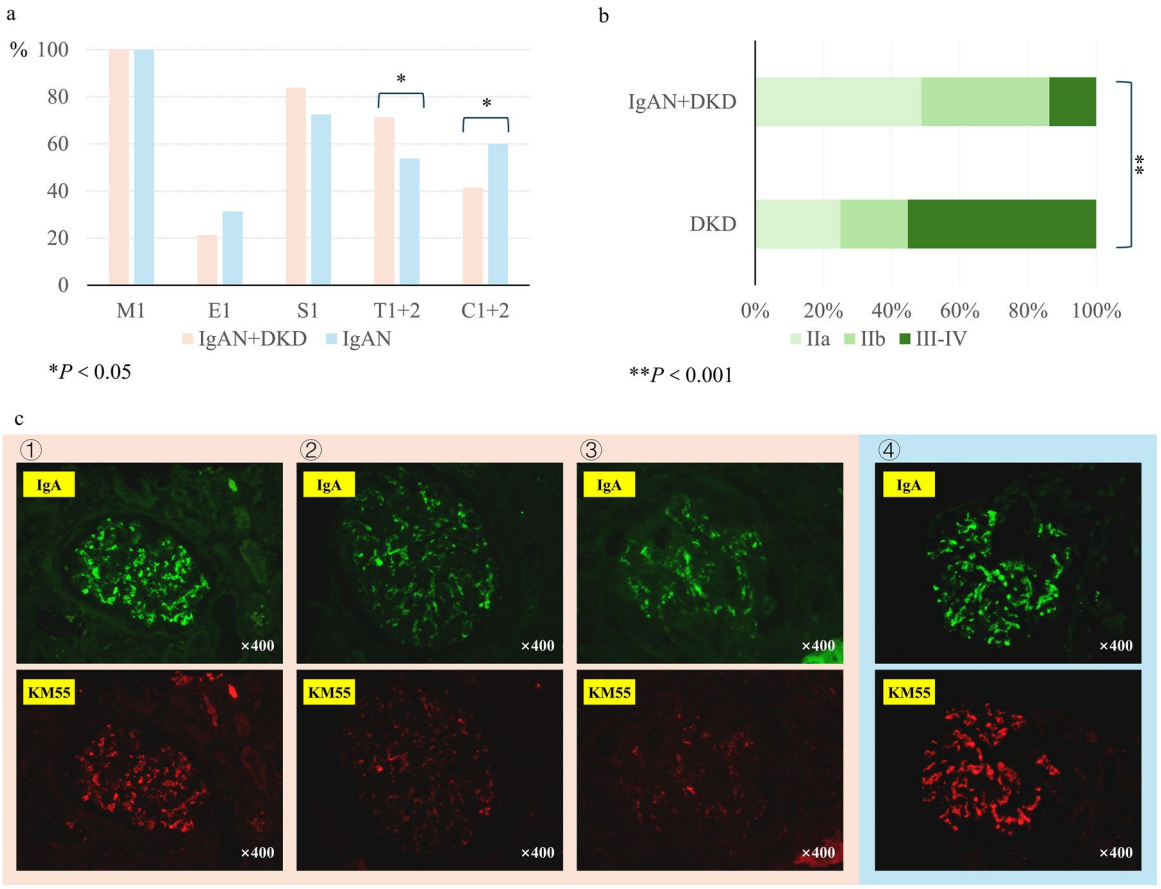


Figure 2. Renal pathological grading and immunofluorescence of KM55/IgA. (a) Oxford classification of IgAN [12]. M1: >50% of glomeruli show increased cells. E1: endocapillary hypercellularity; S1: scarring in parts of the glomeruli; T1 + 2: >25% damage in tubular atrophy/interstitial fibrosis; C1 + 2: crescents in Bowman's space. Tubulointerstitial lesions were more severe in IgAN + DKD group than in IgAN group (71.25% vs. 53.75%, $p = 0.022$). The proportion of crescent formation in the IgAN + DKD group was lower than that in the IgAN group (41.25% vs. 60%, $p = 0.018$). (b) Glomerular classification of diabetic nephropathy [13]. IIa: mild mesangial expansion (mesangium ≤ capillary lumen) in >25% of glomeruli; IIb: severe mesangial expansion (mesangium > capillary lumen) in >25% of glomeruli; III–IV: presence of Kimmelstiel–Wilson nodules or global glomerulosclerosis affecting >50% of glomeruli. Glomerular lesions in the IgAN + DKD group were milder than those in the DKD group ($p < 0.001$). (c) Immunofluorescence of KM55/IgA: ① No. 14185. HKY, M/35 years, IgAN + DKD. 24h-uPro 0.76 g, uRBC 95/μL, sCr 118 μmol/L, IF: IgA 3+, KM55 2+; ② No. 21814. PLH, F/56 years, IgAN + DKD. 24h-uPro 2.09 g, uRBC 1000/μL, sCr 131 μmol/L, IF: IgA 3+, KM55 +; ③ No. 21992. JYQ, F/54 years, IgAN + DKD. 24h-uPro 1.31 g, uRBC 38/μL, sCr 69 μmol/L, IF: IgA 2+ to 3+, KM55 1+ to 2+; ④ No. 24231. WXH, F/50 years, IgAN. 24h-uPro 0.15 g, uRBC 80/μL, sCr 70 μmol/L, IF: IgA 3+, KM55 3+. KM55: anti-human Gd-IgA1; IgAN: IgA nephropathy; DKD: diabetic kidney disease; 24h-uPro: 24-hour urinary protein; uRBC: urine red blood cell; sCr: serum creatinine; IF: immunofluorescence.

difference in tubulointerstitial lesions, but arteriole sclerosis was more severe in the DKD group (Supplementary Tables 2-3). IgA deposition was more prominent in the IgAN + DKD group, with similarly increased deposition of IgM, C3, λ and κ (Supplementary Table 4).

3.3. Treatment and follow-up

All the groups received supportive care. ACEI/ARB and SGLT2i were administered based on eGFR and tolerability. The proportion of immunosuppressive therapy was comparable between IgAN + DKD and IgAN (80% vs. 85%, $p = 0.405$) but higher than that in DKD (61.25%, $p = 0.009$). The immunosuppressive therapy in the DKD group was TW.

We collected follow-up data at 12 ± 0.5 months post-biopsy to assess therapeutic outcomes (Table 2). Only 31 IgAN + DKD patients, 41 IgAN patients and 44 DKD patients had available 1-year data and were included in the complete-case analysis. Attrition was relatively high (61.25% in the IgAN + DKD group, 48.75% in the IgAN group and 45% in the DKD group), which may introduce attrition bias. To evaluate whether attrition led to systematic bias, we performed within-group comparison of baseline characteristics between participants with complete follow-up (non-LTFU) and those lost to follow-up (LTFU), showing inter-group differences in terms of sex, age, Hb level and the proportion of MH (Supplementary Table 6). Binary logistic regression models with LTFU as the outcome indicated that in both IgAN + DKD and

Table 2. Treatment and follow-up.

	IgAN + DKD	IgAN	DKD	*p* (IgAN + DKD vs. IgAN)	*p* (IgAN + DKD vs. DKD)
Treatment					
ACEI/ARB/SGLT2i, *n* (%)	70 (87.50)	69 (86.25)	64 (80.00)	0.815	0.199
Glucocorticoids/Immunosuppressant, *n* (%)	64 (80.00)	68 (85.00)	49 (61.25)	0.405	0.009
Lipid regulating agents, *n* (%)	49 (61.25)	32 (40)	54 (67.50)	0.007	0.409
Follow-up time, months	16.57 (2.88, 37.47)	25.40 (4.27, 44.07)	19.35 (2.37, 42.37)	0.326	0.685
Endpoints, *n* (%)	15 (18.75)	5 (6.25)	31 (38.75)	0.017	0.005
Number of patients followed up for 6-month, *n* (%)	49/80 (61.25)	54/80 (67.50)	52/80 (65.00)		
24h-uPro 6-month rate of change (%)	−44.73 (−69.76, −17.58)	−62.16 (−78.67, −26.67)	−14.36 (−47.21, 21.47)	0.118	0.031
eGFR 6-month rate of change (%)	7.20 (−8.43, 33.16)	10.54 (−1.28, 35.36)	−1.76 (−14.62, 12.46)	0.315	0.034
6-Month clinical remission status, *n* (%)				0.040	0.022
Complete response	6/49 (12.25)	18/54 (33.33)	7/52 (13.46)		
Partial response	26/49 (53.06)	21/54 (38.89)	14/52 (26.92)		
No response	17/49 (34.69)	15/54 (27.78)	31/52 (59.62)		
Number of patients followed up for 1-year, *n* (%)	31/80 (38.75)	41/80 (51.25)	44/80 (55.00)		
24h-uPro 1-year rate of change (%)	−57.65 (−80.57, −13.74)	−69.90 (−87.89, −30.99)	−35.73 (−67.52, 20.72)	0.073	0.116
eGFR 1-year rate of change (%)	1.42 (−6.91, 12.53)	3.66 (−2.82, 16.06)	−4.47 (−23.35, 8.24)	0.237	0.08
TC 1-year (mmol/L)	4.35 (3.85, 5.12)	4.29 (3.55, 5.26)	4.81 (4.24, 6.17)	0.978	0.039
TG 1-year (mmol/L)	1.21 (0.79, 1.64)	1.41 (0.92, 1.76)	1.45 (0.91, 2.03)	0.505	0.397
LDL-C 1-year (mmol/L)	2.31 (2.12, 2.76)	2.67 (1.99, 3.41)	2.48 (1.76, 3.84)	0.299	0.391
HDL-C 1-year (mmol/L)	1.14 (0.92, 1.29)	1.13 (1.00, 1.37)	1.25 (0.97, 1.54)	0.490	0.247
1-Year clinical remission status, *n* (%)				0.003	0.186
Complete response	4/31 (12.90)	21/41 (51.22)	7/44 (15.91)		
Partial response	14/31 (45.16)	11/41 (26.83)	11/44 (25.00)		
No response	13/31 (41.94)	9/41 (21.95)	26/44 (59.09)		
eGFR annual decline (mL/min/1.73 m^2)	−0.71 (−1.06, 0.64)	−0.09 (−0.67, 1.22)	−3.63 (−6.44, −0.98)	0.240	0.003

Basic treatment: low-salt diet, blood pressure control and blood glucose control; ACEI: angiotensin-converting enzyme inhibitors; ARB: angiotensin II receptor blockers; SGLT2i: sodium-glucose transporter 2 inhibitors; ACEI/ARB and SGLT2i were used according to GFR and adverse drug reactions; 24h-uPro: 24-hour urinary protein; eGFR: estimated GFR; TC: total cholesterol; TG: triglyceride; LDL-C: low-density lipoprotein cholesterol; HDL-C: high-density lipoprotein cholesterol.

Endpoints: a permanent decline in eGFR of ≥50% from baseline or entry into kidney failure (eGFR <15 mL/min/1.73 m^2 or maintenance dialysis); clinical remission status: complete response, defined as 24h-uPro ≤0.3 g, and sCr increase less than 10% compared to baseline; partial response, defined as 0.3 g < 24h-uPro < 1 g with a reduction of ≥50% from baseline, and sCr increase less than 10% compared to baseline; non-response, defined as 24h-uPro ≥0.3 g with a reduction of <50% or no reduction from baseline, or sCr increase of more than 10% compared to baseline. 24h-uPro 1-year rate of change (%) = [(1-year 24h-uPro) − (baseline 24h-uPro)]/(baseline 24h-uPro) × 100%; eGFR 1-year rate of change (%) = (1-year eGFR − baseline eGFR)/baseline eGFR × 100%; eGFR annual decline (mL/min/1.73 m^2) = (last follow-up time eGFR − baseline eGFR)/follow-up time.

IgAN groups, receipt of immunosuppressive therapy was associated with lower risk of LTFU; in the IgAN + DKD group, older age increased the risk of attrition; in the DKD group, higher haemoglobin was associated with higher attrition risk (Supplementary Table 7). Missing pattern analysis revealed mostly non-significant differences between non-LTFU and LTFU groups, with only a few variables showing weak or marginally significant associations; (the age of the group with a 1-year missing of 24h-uPro was nearly significantly higher than that of the group with it ($t = -1.9$, $df = 58.0$, $p = 0.062$); the diabetes duration of the group with a last missing of eGFR was lower than that of the group with it ($t = 1.8$, $df = 25.3$, $p = 0.084$). Little's MCAR test yielded non-significant p values in all three groups (IgAN + DKD $\chi^2 = 126.585$, $df = 139$, $p = 0.767$; IgAN $\chi^2 = 20.552$, $df = 18$, $p = 0.303$; DKD $\chi^2 = 101.520$, $df = 107$, $p = 0.631$), failing to reject the MCAR assumption. However, given the high attrition rates and the finding that immunosuppressive therapy was associated with a lower risk of LTFU, the missing data mechanism may not be purely MCAR. Therefore, although complete-case analysis was performed, the 12-month remission estimates may still be affected by attrition bias, potentially reducing precision and generalizability. Sensitivity analyses were not feasible due to the limited sample size after attrition; future studies with more complete follow-up are warranted to confirm these findings.

Clinical efficacy was evaluated based on 24h-uPro and eGFR in the complete-case subset (Table 2, Figure 3(a)). Compared to baseline, IgAN + DKD group exhibited a reduction in 24h-uPro (−1.65 (−2.22, −1.08) g, $p < 0.001$), corresponding to a 57.65% decline. Although this reduction was greater than that in the DKD group (35.73%) and less than that in the IgAN group (69.90%), intergroup comparisons did not reach statistical significance. In IgAN + DKD group, eGFR showed a non-significant increase from baseline (0.17 (−5.64, 5.98) mL/min/1.73 m^2, $p = 0.954$), corresponding to a 1.42% change. This increment was numerically lower than that in the IgAN group (3.66%) and contrasted with the decline observed in the DKD group (−4.47%). However, the difference in eGFR percentage change among the three groups did not reach statistical significance. The annual decline in eGFR in the IgAN + DKD group was lower than that in the DKD group (−0.71 mL/min/1.73 m^2 vs. −3.63 mL/min/1.73 m^2, $p = 0.003$), while with no difference compared to IgAN group (−0.71 mL/min/1.73 m^2 vs. −0.09 mL/min/1.73 m^2, $p = 0.240$).

In the complete-case analysis, at 12 ± 0.5 months post-biopsy, the overall clinical remission rate in the IgAN + DKD group was 58.06% (18/31), which was not statistically significantly different from that in the IgAN group (78.05% (32/41), $p = 0.078$) and the DKD group (40.91% (18/44), $p = 0.165$). And the complete response rate of the IgAN + DKD group was lower than the IgAN group (12.9% (4/31) vs. 51.22% (21/41), $p = 0.001$). In the IgAN + DKD group, the remission group showed a significantly higher crescent ratio (8.33% vs. 0.00%, $p < 0.001$) as well as the usage rate of glucocorticoids/immunosuppressants (100.00% (18/18) vs. 69.23% (9/13), $p = 0.023$). The prevalence of capillary endothelial cell proliferation (38.89% (7/18) vs. 7.69% (1/13), $p = 0.095$) showed a borderline trend towards association with remission. Given the near-perfect separation of immunosuppressant use across the two outcome groups, this variable was excluded from the multivariable regression models to avoid complete data separation and unstable coefficient estimation. Progressive nested multivariable binary logistic regression was performed to adjust for potential confounders (Supplementary Table 8). In this exploratory multivariable analysis, capillary endothelial cell proliferation was associated with higher odds of clinical remission (aOR = 15.11, 95%CI 1.04–220.11, $p = 0.047$); the extremely wide CI indicated substantial uncertainty. Crescent ratio only showed a borderline positive association with higher likelihood of remission (aOR = 1.32, 95%CI 0.96–1.81, $p = 0.088$) (Supplementary Table 9). In this 1-year follow-up cohort, only four patients did not receive glucocorticoid/immunosuppressive therapy, none of whom achieved clinical remission. Given the extremely small sample size in the non-immunosuppression group, formal between-group statistical comparisons were not performed.

The median survival time of IgAN + DKD group was 65.63 months (95%CI 35.19–96.07). By the end of 2024, a total of 15 patients reached the composite endpoint, while 65 did not. Five IgAN patients reached the composite endpoint (median survival time could not be calculated). A statistically significant difference was observed in the cumulative survival-free of adverse renal endpoint events between the IgAN + DKD and IgAN group (HR = 2.821, 95%CI 1.174–6.781, $p = 0.032$). In the DKD group, 31 endpoint events occurred, with a median survival time of 55.23 months (95%CI 30.52–79.94). There was a difference in the cumulative survival rate between the IgAN + DKD group and DKD group (HR = 0.525, 95%CI 0.293–0.943, $p = 0.028$). Overall, the risk of renal function deterioration in the IgAN + DKD group was

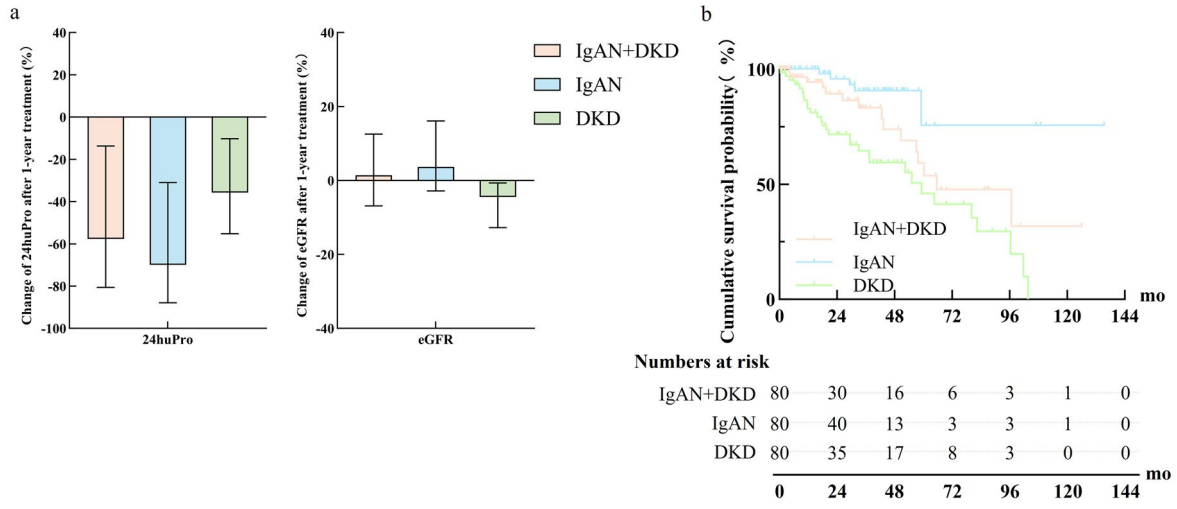


Figure 3. Change of 24h-uPro, eGFR after 1-year treatment and cumulative survival probability. (a) Change of 24h-uPro, eGFR after 1-year treatment (complete-case analysis). Data were presented as median (interquartile range); error bars represented interquartile range. The 1-year rate of change in 24h-uPro and eGFR showed no statistically significant differences between the IgAN + DKD group and either the IgAN or DKD groups. (b) Cumulative survival probability free of the composite renal endpoints (Kaplan–Meier's curves with log-rank test). Numbers at risk were displayed below the *x*-axis at each time point. The risk of renal function deterioration in the IgAN + DKD group was higher than that in the IgAN group (HR = 2.821, 95%CI 1.174–6.781, $p = 0.032$) but lower than that in the DKD group (HR = 0.525, 95%CI 0.293–0.943, $p = 0.028$). 24h-uPro: 24-hour urinary protein; eGFR: estimated glomerular filtration rate; IgAN: IgA nephropathy; DKD: diabetic kidney disease.

higher than that in the IgAN group but lower than that in the DKD group (Figure 3(b)). In the IgAN + DKD group, compared with those who did not reach the renal endpoint, patients who reached the endpoint were significantly older, had lower Hb levels, higher prevalence of DR, and higher 24h-uPro excretion. Pathologically, the endpoint group demonstrated greater GBM thickening and more severe interstitial inflammatory cell infiltration. Notably, the prevalence of capillary endothelial cell proliferation was markedly lower in the endpoint group (0.00% (0/15) vs. 26.15% (17/65), $p = 0.032$). Although the proportion of patients treated with glucocorticoids/immunosuppressants was higher in the non-endpoint group than that in the endpoint group, this difference did not reach statistical significance (81.54% vs. 73.33%, $p = 0.485$) (Supplementary Tables 10, and 11). In the pre-specified exploratory analyses, univariate Cox regression analysis identified several factors associated with the composite endpoint in patients with IgAN + DKD, including 24h-uPro, sCr, DR, HbA1c, Hb, sC3 and GBM thickness. In the subsequent exploratory multivariate Cox regression model, baseline sCr (HR = 1.023, 95%CI 1.006–1.040, $p = 0.008$) and GBM thickness (HR = 1.007, 95%CI 1.000–1.013, $p = 0.039$) were associated with the endpoint. Limited by the low number of events and the attendant risk of overfitting, these findings should be interpreted with caution and do not support claims of independent prognostic prediction (Supplementary Table 12).

In addition, although IgAN group exhibited pronounced lipid metabolism disorders than IgAN + DKD group at baseline, lipid levels in IgAN group were mostly controlled within the normal range after 1-year of treatment. Notably, the utilization rate of lipid-lowering agents in the IgAN group was lower, indicating that the fluctuations in lipid levels in the IgAN group may be related to immune disorders to a greater extent (Table 2).

4. Discussion

In our study, IgAN + DKD was more prevalent in males (72.50%) with a mean age of 53.86 years at the time of renal biopsy. The ratio of IgAN + DKD to IgAN among patients undergoing renal biopsy during the same period was approximately 1:40, consistent with the findings reported by Jiang et al. [3]. In the unadjusted baseline comparison, the IgAN + DKD group exhibited higher baseline levels of 24h-uPro and sCr, as well as a lower incidence of MH, compared with the IgAN group. Additionally, MH was more

prominent in patients with IgAN + DKD than in those with DKD. After PSM (1:1, matching factors: sex, age and sCr), the statistically significant between-group differences in 24h-uPro and MH remained stable across all three groups.

The differences in proteinuria and haematuria among the three groups can be partially explained by renal pathological features. Proteinuria is mainly attributed to impairment of the glomerular filtration barrier and reabsorption dysfunction of tubular epithelial cells. Mesangial matrix expansion is common in both IgAN and DKD. In IgAN, this phenomenon is mainly driven by the deposition of Gd-IgA1 in the mesangium [16], whereas in DKD, it's initiated with hyperglycaemic stimulation [17]. Mesangial matrix expansion squeezes capillary loops, destroying glomerular filtration barrier and reducing blood flow, eventually resulting in glomerulosclerosis [18]. GBM thickening results from the accumulation of advanced glycosylation end products, the activation of the inflammatory signalling pathway, and oxidative stress damage, which destroy normal structure of filtration membrane [19,20]. Podocytes and their foot processes form a critical component of the glomerular filtration barrier and are essential to maintain renal filtration selectivity and prevent protein leakage [21]. Fusion or even disappearance of foot processes can lead to proteinuria [22]. In DKD, impaired glycolysis enhances ornithine decomposition, and then disturb cytoskeletal remodelling, resulting in foot process fusion [23]. Mesangial matrix expansion, GBM thickening and foot process fusion are positively correlated with proteinuria level [24–27]. In this study, IgAN + DKD exhibited more severe mesangial matrix expansion, GBM thickening compared with IgAN group, consistent with the observed differences in 24h-uPro. Although patients with IgAN + DKD demonstrated a more pronounced degree of foot process fusion, there was no statistically significant difference in 24h-uPro between the IgAN + DKD group and the DKD group, which may be explained by the milder mesangial matrix expansion and GBM thickening in the IgAN + DKD group. Tubular epithelial cells are responsible for reabsorbing most of the small-molecule proteins filtered by the glomerulus. When the renal tubule atrophied, the tight junctions of tubule epithelial cells were destroyed and transformed into a fibrotic phenotype. The reabsorption function decreases, the small molecular protein in urine increase [28]. The proportion of patients with >25% renal tubular atrophy in IgAN + DKD was higher than that in IgAN, which may be one of the factors contributing to elevated 24h-uPro in IgAN + DKD.

Glomerular haematuria is a feature of IgAN and rarely observed in patients with DKD. In IgAN, deposition of Gd-IgA1 recruits inflammatory cells, triggering complement system activation and crescent formation. These immune-inflammatory responses directly impair the glomerular filtration barrier, creating gaps that permit red blood cells to leak into the urine [7,29,30]. In DKD, the GBM thickens progressively under hyperglycaemia but does not rupture, thereby preventing extravasation of red blood cells. We found that the incidence of haematuria, uRBC count, and proportion of dysmorphic uRBC in IgAN + DKD patients were lower than those in IgAN patients. This might be related to less deposition of immune substances (λ, κ), less activation of the complement system (C3), less crescent formation and thicker GBM in IgAN + DKD.

KM55, a monoclonal antibody that specifically recognizes the aberrantly galactosylated O-glycan epitope on Gd-IgA1, has been validated as a sensitive probe to detect Gd-IgA1 deposits in the mesangium of IgAN patients. KM55 reactivity co-localizes precisely with mesangial Gd-IgA1 deposits in renal biopsy specimens [31]. The assembled Gd-IgA1-containing immune complexes trigger complement activation predominantly via the lectin and alternative pathways, driving the sequential formation of C3 convertase and membrane attack complex. This complement cascade promotes mesangial cell proliferation, podocyte injury and contributes to the progressive renal damage [9]. In previous studies, we demonstrated that in IgA-depositing kidney diseases secondary to systemic disorders or infections, KM55 staining is negative or weaker, and the KM55/IgA ratio is lower [15]. Extending these earlier observations, the current study shows that the KM55 fluorescence intensity in the IgAN + DKD group remains comparable to that in the IgAN group, while the KM55/IgA intensity ratio is statistically significantly reduced. This finding supports the notion that the KM55/IgA ratio represents a potentially valuable histopathological marker for distinguishing primary IgAN and IgAN + DKD. Although we observed glomerular C3 immunofluorescence intensity was significantly lower in the IgAN + DKD group than in the IgAN group, this single histological marker alone is insufficient to fully quantify and compare the overall degree of intrarenal complement activation between the two groups.

A low-salt diet, blood pressure and glucose control, ACEI/ARB, and SGLT2i constitute the basic therapeutic approaches for both IgAN and DKD [9–11]. Immunosuppressive therapy is indicated for IgAN with

persistent 24h-uPro ≥1 g pathological active lesions (e.g. cellular crescents) or a high risk of disease progression. DKD is thought to be a non-immune-mediated renal impairment, and immunosuppressants are generally not administered. However, there is growing recognition that DKD involves renal damage associated with metabolic dysfunction, chronic inflammation and complex immune-mediated interactions [32,33]. Many researches have confirmed that *Tripterygium* glycosides, a weak immunosuppressant, can reduce 24h-uPro of DKD [34]. In our study, 1-year clinical outcomes across the three groups were evaluated using changes in 24h-uPro and eGFR in complete-case analyses, with no statistically significant intergroup difference observed. We further assessed clinical remission rates: while overall clinical remission showed no significant difference across groups, the rate of complete response was significantly lower in the IgAN + DKD group than in the IgAN group. In exploratory multivariable analyses, capillary endothelial cell proliferation was associated with clinical remission in the IgAN + DKD group, while the crescent ratio showed only a borderline positive association. These findings should be interpreted with caution given the limited number of complete cases and wide CI. Although immunosuppressant use was more frequent in the remission subgroup, the limited sample size precluded a definitive conclusion on the therapeutic benefit of immunosuppression for remission induction.

Glomerular injuries, including mesangial matrix expansion, GBM thickening and foot process fusion, along with chronic pathologies such as tubular atrophy, interstitial fibrosis and renal arteriolar sclerosis, contribute to an irreversible decline in renal function [27]. In our study, survival analysis showed that the risk of renal function deterioration in IgAN + DKD was higher than that in IgAN, but lower than that in DKD. Baseline sCr and GBM thickness were significantly associated with the composite endpoint in exploratory analyses. Notably, while capillary endothelial cell proliferation, a classic IgAN marker of active immune injury, was markedly less prevalent in the endpoint subgroup, it was not significantly related to endpoint events in the exploratory analyses. This indicates that in IgAN + DKD, once extensive chronic diabetic glomerular injury, exemplified by irreversible GBM thickening, is established, acute immune-active IgAN-related lesions no longer act as the dominant driver of progression. The proportion of patients receiving immunosuppressive therapy also showed no statistically significant difference between endpoint and non-endpoint subgroups, suggesting that current unselected use of immunosuppressive therapy in this cohort does not produce a measurable modifying effect on long-term renal prognosis.

We also observed meaningful findings in terms of the cardiovascular system and lipid metabolism. The degree of microangiopathy, including DR and renal arteriolar sclerosis, in IgAN + DKD was more severe than that in IgAN, but less severe than that in DKD. Diabetic retinopathy and nephropathy share common pathogenic mechanisms, with substantial evidence implicating oxidative stress, accumulation of advanced glycation end products, and genetic susceptibility as key drivers of both microvascular complications [35]. It had been established the progression of DKD microvascular complications correlated with the duration of diabetes [36]. In this study, the incidence of DR and the severity of renal arteriolar sclerosis in the IgAN + DKD group remained significantly lower than in the DKD group, despite no statistically significant between-group difference in diabetes duration. A plausible mechanistic interpretation is that underlying IgAN-related glomerular immune inflammation partially rewires downstream hyperglycaemia-mediated microvascular signalling pathways, a hypothesis that warrants further experimental validation in dedicated translational studies. Compared with IgAN, patients with IgAN + DKD exhibit an earlier onset of cardiac structural abnormalities, specifically in the left ventricle, along with impaired systolic function. These cardiac alterations might be mediated by a longer duration of hypertension and diabetes as well as a higher BMI. Chronic systemic hypertension induces cardiac remodelling, which is characterized by left ventricular hypertrophy and chamber dilation. The progression from diastolic dysfunction to systolic impairment ultimately culminates in heart failure [37]. Diabetes and obesity drive pathological cardiac remodelling characterized by hypertrophy, systolic dysfunction and diastolic impairment, mediated through fibrotic transformation, chronic inflammation and mitochondrial metabolic derangements [38]. Despite higher BMI and proportion of obesity in IgAN + DKD, IgAN demonstrated more pronounced baseline dyslipidaemia. Following primary immunosuppressive therapy targeting IgAN, rather than lipid-lowering therapy (the usage rate of lipid-lowering agents was lower in the IgAN group than in the IgAN + DKD group), serum lipid levels in IgAN patients normalized within one year. This suggests that dyslipidaemia in IgAN is likely associated with the underlying immune dysregulation characteristic of the disease.

5. Conclusions

Patients with IgAN + DKD manifested milder haematuria and comparable proteinuria versus IgAN. Markers of active lesions, including glomerular immune complex deposition and crescent formation, were less prominent in IgAN + DKD. The KM55/IgA ratio might serve as a potential biomarker for distinguishing IgAN from IgAN + DKD. The overall burden of proliferative lesions in IgAN + DKD patients was milder than that in DKD, with lower severity of mesangial matrix expansion, fewer K–W nodules, less extensive GBM thickening and less severe renal arteriosclerosis. Renal progression risk in IgAN + DKD was higher than in IgAN but lower than in DKD. Baseline sCr and irreversible GBM thickening were linked to the endpoint in exploratory analyses. Unselected immunosuppressive therapy showed no measurable long-term renal benefit in this cohort, indicating the necessity of pre-treatment pathological stratification for this understudied population – a strategy that remains to be validated in larger prospective cohorts and dedicated interventional trials.

Our study had several limitations. This was a retrospective and single-centre study, which may restrict the generalizability of the findings. Our biopsy-based cohort is subject to selection bias, as kidney biopsies are not performed universally but are reserved for cases with clinical features warranting histological confirmation. This may result in over-representation of severe, progressive or atypical disease, potentially limiting external validity and exaggerating associations in more advanced cases. Pathology diagnoses were made in routine clinical practice without formal blinding to clinical group, which may introduce some observer bias, although group assignment for the study was based on predefined strict criteria. Missing 1-year response data in 45–61.25% of cases introduces potential selection bias. The relatively short follow-up duration may preclude definitive conclusions regarding long-term outcomes. Another key limitation is the potential for confounding by indication, whereby patients receiving immunosuppression may have had more severe or rapidly progressive disease prompting treatment, which could bias associations towards the null. Although we adjusted for multiple measured confounders, residual confounding from unmeasured variables remains possible and could influence our findings. To more accurately delineate the clinical, pathological and prognostic distinctions of IgAN superimposed on DKD, future multicentre prospective studies with larger cohorts are warranted.

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Ethics statement

This study was approved by the Ethics Committee of Hangzhou Traditional Chinese Medicine Hospital (HZTCMH) (Approval No. 2025KLL010).

Consent form

The requirement for consent was waived by the Ethics Committee.

Author contributions

CRediT: **Anni Wang**: Data curation, Writing – original draft; **Wen Shi**: Data curation, Writing – original draft; **Kunyue Xu**: Formal analysis, Software; **Yuan Yuan**: Data curation; **Xuanli Tang**: Conceptualization; **Shuman Zhao**: Data curation; **Dongrong Yu**: Conceptualization; **Hong Liu**: Formal analysis, Software; **Xiang Liu**: Formal analysis, Software; **Xue Jiang**: Conceptualization.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

Original data generated and analysed during this study are included in this published article. All remaining data are available from the corresponding author upon reasonable request.

References

[1] Selvaskandan H, Barratt J, Cheung CK. Novel treatment paradigms: primary IgA nephropathy. Kidney Int Rep. 2024;9(2):203–213. doi: 10.1016/j.ekir.2023.11.026.
[2] Wong K, Pitcher D, Braddon F, et al. Effects of rare kidney diseases on kidney failure: a longitudinal analysis of the UK National Registry of Rare Kidney Diseases (RaDaR) cohort. Lancet. 2024;403(10433):1279–1289. doi: 10.1016/S0140-6736(23)02843-X.
[3] Jiang L, Dong B, Yan Y, et al. Clinicopathological analysis of IgA nephropathy combined with other glomerular diseases. Medicine. 2019;98(41):e17388. doi: 10.1097/MD.0000000000017388.
[4] Dasgupta I, Zac-Varghese S, Chaudhry K, et al. Current management of chronic kidney disease in type-2 diabetes—a tiered approach: an overview of the Joint Association of British Clinical Diabetologists and UK Kidney Association (ABCD-UKKA) guidelines. Diabet Med. 2025;42(2):e15450. doi: 10.1111/dme.15450.
[5] He X, Deng Y, Tian B, et al. A retrospective cohort study of clinical characteristics and outcomes of type 2 diabetic patients with kidney disease. PeerJ. 2024;12:e16915. doi: 10.7717/peerj.16915.
[6] Kritmetapak K, Anutrakulchai S, Pongchaiyakul C, et al. Clinical and pathological characteristics of non-diabetic renal disease in type 2 diabetes patients. Clin Kidney J. 2018;11(3):342–347. doi: 10.1093/ckj/sfx111.
[7] Nihei Y, Suzuki H, Suzuki Y. Current understanding of IgA antibodies in the pathogenesis of IgA nephropathy. Front Immunol. 2023;14:1165394. doi: 10.3389/fimmu.2023.1165394.
[8] Thomas MC, Brownlee M, Susztak K, et al. Diabetic kidney disease. Nat Rev Dis Primers. 2015;1(1):15018. doi: 10.1038/nrdp.2015.18.
[9] Gleeson PJ, O'Shaughnessy MM, Barratt J. IgA nephropathy in adults—treatment standard. Nephrol Dial Transplant. 2023;38(11):2464–2473. doi: 10.1093/ndt/gfad146.
[10] Filippone EJ, Gulati R, Farber JL. Contemporary review of IgA nephropathy. Front Immunol. 2024;15:1436923. doi: 10.3389/fimmu.2024.1436923.
[11] Liew A, Bavanandan S, Hao CM, et al. Executive summary of the Asian Pacific Society of nephrology clinical practice guideline on diabetic kidney disease-2025 update. Nephrology. 2025;30(5):e70031. doi: 10.1111/nep.70031.
[12] Chen S, Chen L, Jiang H. Prognosis and risk factors of chronic kidney disease progression in patients with diabetic kidney disease and non-diabetic kidney disease: a prospective cohort CKD-ROUTE study. Ren Fail. 2022;44(1):1309–1318. doi: 10.1080/0886022X.2022.2106872.
[13] Trimarchi H, Barratt J, Cattran DC, et al. Oxford classification of IgA nephropathy 2016: an update from the IgA Nephropathy Classification Working Group. Kidney Int. 2017;91(5):1014–1021. doi: 10.1016/j.kint.2017.02.003.
[14] Tervaert TW, Mooyaart AL, Amann K, et al. Pathologic classification of diabetic nephropathy. J Am Soc Nephrol. 2010;21(4):556–563. doi: 10.1681/ASN.2010010010.
[15] Tang X, Zheng J, Jiang X, et al. The significance of galactose-deficient immunoglobulin A1 staining in kidney diseases with IgA deposition. Int Urol Nephrol. 2023;55(8):2119–2129. doi: 10.1007/s11255-023-03512-5.
[16] Boi R, Ebefors K, Nyström J. The role of the mesangium in glomerular function. Acta Physiol. 2023;239(2):e14045. doi: 10.1111/apha.14045.
[17] Dwivedi S, Sikarwar MS. Diabetic nephropathy: pathogenesis, mechanisms, and therapeutic strategies. Horm Metab Res. 2025;57(1):7–17. doi: 10.1055/a-2435-8264.
[18] Singh J, Jain A, Bhamra R, et al. The mechanistic role of different mediators in the pathophysiology of nephropathy: a review. Curr Drug Targets. 2023;24(2):104–117. doi: 10.2174/1389450124666221026152647.

[19] Huang X, Li B, Hu J, et al. Advanced glycation endproducts mediate chronic kidney injury with characteristic patterns in different stages. Front Physiol. 2022;13:977247. doi: 10.3389/fphys.2022.977247.
[20] Wang N, Zhang C. Oxidative stress: a culprit in the progression of diabetic kidney disease. Antioxidants. 2024;13(4):455. doi: 10.3390/antiox13040455.
[21] Meliambro K, He JC, Campbell KN. Podocyte-targeted therapies – progress and future directions. Nat Rev Nephrol. 2024;20(10):643–658. doi: 10.1038/s41581-024-00843-z.
[22] Wang Y, Yu X, Wang Z, et al. Mitochondrial morphological aberrations and redistribution in podocytes with glomerular diseases. Ren Fail. 2025;47(1):2515528. doi: 10.1080/0886022X.2025.2515528.
[23] Luo Q, Liang W, Zhang Z, et al. Compromised glycolysis contributes to foot process fusion of podocytes in diabetic kidney disease: role of ornithine catabolism. Metabolism. 2022;134:155245. doi: 10.1016/j.metabol.2022.155245.
[24] Yan M, Liu S, Zhang M, et al. Mesangial cell-derived tenascin-C contributes to mesangial cell proliferation and matrix protein production in IgA nephropathy. Nephrology. 2022;27(5):458–466. doi: 10.1111/nep.14031.
[25] Zang L, Saitoh S, Katayama K, et al. A zebrafish model of diabetic nephropathy shows hyperglycemia, proteinuria and activation of the PI3K/Akt pathway. Dis Model Mech. 2024;17(5):dmm050438. doi: 10.1242/dmm.050438.
[26] Lee JH, Jang SH, Cho NJ, et al. Severity of foot process effacement is associated with proteinuria in patients with IgA nephropathy. Kidney Res Clin Pract. 2020;39(3):295–304. doi: 10.23876/j.krcp.20.017.
[27] Giannou P, Gakiopoulou H, Stambolliu E, et al. Urine nephrin and podocalyxin reflecting podocyte damage and severity of kidney disease in various glomerular diseases—a cross-sectional study. J Clin Med. 2024;13(12):3432. doi: 10.3390/jcm13123432.
[28] Duan ZY, Bu R, Liang S, et al. Urinary miR-185-5p is a biomarker of renal tubulointerstitial fibrosis in IgA nephropathy. Front Immunol. 2024;15:1326026. doi: 10.3389/fimmu.2024.1326026.
[29] Liu Y, Gong Y, Xu G. The role of mononuclear phagocyte system in IgA nephropathy: pathogenesis and prognosis. Front Immunol. 2023;14:1192941. doi: 10.3389/fimmu.2023.1192941.
[30] Yu L, Zhang H, Wu Y. Association between different proportions of crescents and the progression of IgA nephropathy (IgAN): a systematic review and meta-analysis. BMC Nephrol. 2024;25(1):396. doi: 10.1186/s12882-024-03839-w.
[31] Yasutake J, Suzuki Y, Suzuki H, et al. Novel lectin-independent approach to detect galactose-deficient IgA1 in IgA nephropathy. Nephrol Dial Transplant. 2015;30(8):1315–1321. doi: 10.1093/ndt/gfv221.
[32] Mohandes S, Doke T, Hu H, et al. Molecular pathways that drive diabetic kidney disease. J Clin Invest. 2023;133(4):e165654. doi: 10.1172/JCI165654.
[33] Ma L, Liu D, Yu Y, et al. Immune-mediated renal injury in diabetic kidney disease: from mechanisms to therapy. Front Immunol. 2025;16:1587806. doi: 10.3389/fimmu.2025.1587806.
[34] Yan X, Shi J, Zhang Y, et al. Effectiveness and safety of *Tripterygium wilfordii* poly-glycosides on glomerulonephritis: a systematic review and meta-analysis. Front Pharmacol. 2024;15:1339153. doi: 10.3389/fphar.2024.1339153.
[35] Li Y, Wang Q, Chen X, et al. Validation and comparison of diabetic retinopathy-based diagnostic models for diabetic nephropathy. Nan Fang Yi Ke Da Xue Xue Bao. 2023;43(9):1585–1590. doi: 10.12122/j.issn.1673-4254.2023.09.16.
[36] Hang X, Ma J, Wei Y, et al. Renal microcirculation and mechanisms in diabetic kidney disease. Front Endocrinol. 2025;16:1580608. doi: 10.3389/fendo.2025.1580608.
[37] Jiang J, Hu M, Wang Q, et al. Combination of cardiac color Doppler ultrasound, serum MR-ProANP and NT-ProBNP forecasted hypertensive LVH and LFH. Biotechnol Genet Eng Rev. 2024;40(3):2643–2657. doi: 10.1080/02648725.2023.2200626.
[38] Huang S, Chen Z, Li H, et al. High-fat diet-induced diabetic cardiomyopathy in female zebrafish: cardiac pathology and functional decline mediated by type 2 diabetes. Nutrients. 2025;17(13):2209. doi: 10.3390/nu17132209.